

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [189, 155, 203, 171, 207, 173, 187, 194, 141, 170]

Tap amplitudes (pixels): [146.40013661195812, 127.00393694685216, 117.00427342623003, 127.7810627597063, 131.36590120727678, 141.65097952361643, 117.92370414806346, 122.44182292011173, 120.0, 126.21014222319853, 118.92854997854805]

GPT-5

MDS-UPDRS Item 3.4 (Finger Tapping) Scoring Analysis

Interpretation notes:

- The subject was instructed to perform 10 taps; features show:
 - Number of taps detected: 11 (suggests an extra partial tap was captured; periods list has 10 intervals, consistent with 11 peaks).
 - Periods (proxy for speed/rhythm across 10 intervals): [189, 155, 203, 171, 207, 173, 187, 194, 141, 170] frames.
 - Amplitudes (11 tap peaks): [146.4, 127.0, 117.0, 127.8, 131.4, 141.7, 117.9, 122.4, 120.0, 126.2, 118.9] pixels.

Assessing the criteria:

1) Interruptions/hesitations/halts

- Period variability suggests rhythm irregularities. The max period is 207 and min is 141 frames; range = 66 frames.
- Looking for clear interruptions (abnormally long period or large outliers) or a "longer arrest." Here, periods fluctuate but none are extreme outliers relative to neighbors, and there's no single period that stands out as a clear freeze.
- Conclusion: Some rhythm irregularity but not clearly >5 interruptions or a definite long arrest.

2) Speed (slowing)

- Absolute speed can't be directly graded without a calibrated reference. We judge relative consistency:
 - Periods oscillate but stay within a moderate band (141–207 frames), suggesting at most mild-to-moderate slowing compared to expected “as fast as possible.” Without a normative baseline, classify this as slight-to-mild slowing rather than moderate.

3) Amplitude and decrementing pattern

- Check for decrement:
 - Start amplitude ≈ 146.4 (tap 1).
 - Subsequent taps mostly in 117–142 range; several are notably lower than the first.
 - To classify per criteria:
 - “Decrements near the end” (score 1).
 - “Decrements midway through the 10-tap sequence” (score 2).
 - “Decrements starting after the 1st tap” (score 3).
- Sequence overview (by tap index): 1:146 → 2:127 → 3:117 → 4:128 → 5:131 → 6:142 → 7:118 → 8:122 → 9:120 → 10:126 → 11:119.
 - There is an immediate drop from tap 1 to taps 2–3, followed by partial recovery (tap 6 ~142), then lower range again. This looks like a decrement that starts early (just after the first tap), not only near the end, and not merely mid-sequence. Even with partial recovery, the general level remains below the first tap early on.
- Conclusion: The amplitude decrement begins after the first tap.

Overall synthesis:

- Interruptions: present as variability but not clearly >5 or a definitive long arrest.
- Speed: slight-to-mild slowing.
- Amplitude: decrement starting after the first tap, which matches the “Moderate” (3) criterion.

Final score:

3
